3. PROJECT DESIGN

EduGenie — Google Gemini Powered Learning Assistant

Architecture, components, AI modules, API design, and interface design

Project Overview

EduGenie is a lightweight AI-powered educational assistant designed to simplify learning through generative AI. It supports question answering, simplified concept explanations, quiz generation, summarization, and personalized learning recommendations. The supplied project document describes a FastAPI backend with an HTML/CSS frontend and cloud/local AI model integration.

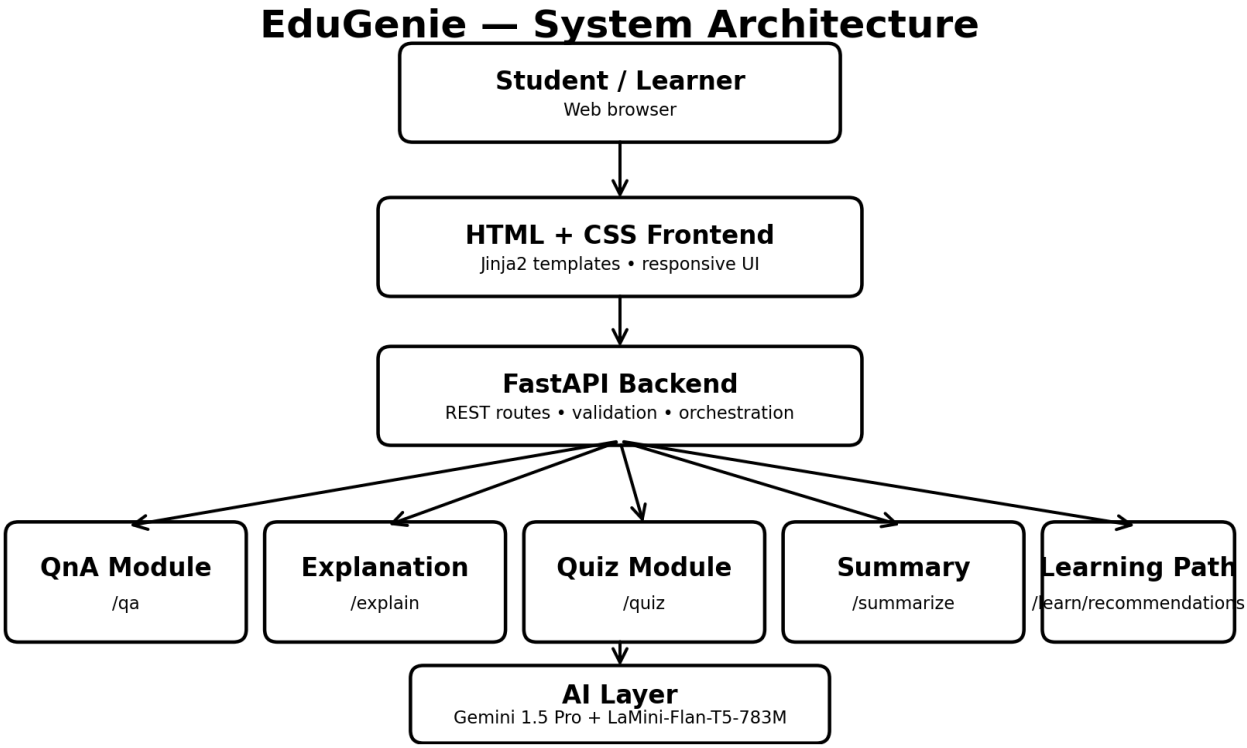


Fig. EduGenie System Architecture

System Architecture

EduGenie follows a lightweight layered web architecture. The learner interacts with the HTML/CSS interface. Form requests are sent to FastAPI routes, which validate inputs and coordinate the corresponding AI module. The application exposes separate endpoints for Q&A;, explanation, quiz generation, summarization, and learning recommendations.

Component Design

Component	Responsibility
main.py	Creates the FastAPI application and coordinates the application entry point.
explanation_module.py	Uses LaMini-Flan-T5-783M for simplified, readable concept explanations.
qna.py	Handles academic and general question-answering through Gemini 1.5 Pro.
quiz_module.py	Generates three MCQs with four options and a correct answer in structured output.
summary_module.py	Summarizes long educational passages into concise, easy-to-understand content.
learning_path.py	Generates structured beginner-to-advanced learning recommendations and resources.
templates/index.html	Provides the user-facing HTML interface and input form.
static/style.css	Provides responsive styling for the web interface.
requirements.txt	Lists Python dependencies required to run the project.

Data / AI Flow

User input → HTML form → FastAPI endpoint → selected AI module → Gemini or local LaMini-Flan-T5 model → processed response → result displayed in the frontend.

API Design

Endpoint	Purpose
/qa	Question answering
/explain	Simplified explanation of a concept
/quiz	Generate a three-question MCQ quiz
/summarize	Summarize educational text
/learn/recommendations	Generate a personalized learning path

AI Module Design

Explanation Module: The project document specifies LaMini-Flan-T5-783M as a lightweight instruction-tuned model for concise, context-aware explanations. It is intended to make complex topics easier for beginners and self-learners to understand.

QnA Module: Gemini 1.5 Pro is used for general knowledge and academic question answering, providing contextual AI-driven assistance across subjects.

Quiz Module: Gemini generates three multiple-choice questions from a supplied passage. Each question has four options and a correct answer. The response is cleaned and parsed as JSON.

Summary Module: Long passages are reduced to concise versions while retaining core information and removing unnecessary repetition.

Learning Path Module: Gemini generates a structured learning path from beginner to advanced concepts, with timelines and learning resources where applicable.

Interface Design

The frontend is described as a simple web interface containing a task selector, text area, submit button, and result container. The task choices include Explain, QnA, Quiz, Summary, and Recommend Path. CSS provides responsive styling for input areas, buttons, and results.

Technology Stack

Layer	Technology	Role
Backend	FastAPI	REST API routing and orchestration
Frontend	HTML + CSS	User interface
Templating	Jinja2	HTML templating
Cloud AI	Google Gemini 1.5 Pro	Q&A, quiz, summary, learning path
Local AI	LaMini-Flan-T5-783M	Concept explanation
Server	Uvicorn	ASGI server
Language	Python 3.10+	Application development

Deployment and Testing

The supplied project document specifies local execution with **uvicorn main:app --reload** and access through **http://127.0.0.1:8000**. Functional testing includes asking questions, obtaining explanations, generating quizzes, summarizing content, and obtaining personalized learning recommendations.

Project Workflow / Milestones

Milestone 1: Model selection and architecture — Gemini 1.5 Pro for cloud tasks and LaMini-Flan-T5-783M for local explanations.

Milestone 2: Core functionality development — implementation of the five AI modules and FastAPI REST endpoints.

Milestone 3: Frontend development — HTML form, CSS styling, and live backend integration.

Milestone 4: Deployment — local execution and functional testing.

Design Reference Note

This project-design PDF is prepared from the supplied EduGenie project document and the supplied Project Design reference. The reference's organization—system architecture, component design, data/flow-oriented design, interface design, and technology quick reference—has been adapted to the EduGenie project without introducing unrelated application components.

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